

Dennis Fang

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EDUCATION

University of California, San Diego

Expected Mar 2027

Master of Science, Data Science | GPA - 4.0/4.0

University of California, Davis

Sept 2021 - Jun 2025

Bachelor of Science, Statistics (Data Science Track) | Minor: Atmospheric Sciences | GPA - 3.7/4.0

- **Coursework:** Machine Learning, Practice in Statistical Data Science, Big Data and High Performance Statistical Computing, Data and Web Technologies for Data Analysis, Advanced Statistical Computing, Statistical Learning, Time Series Analysis

RELEVANT SKILLS

- **Programming and Technical Skills:** Python (Xarray, Pandas, Numpy, Scikit-learn, Statsmodels, Pytorch, Tensorflow), R (ggplot2, tidyverse, caret), SQL
- **Tools and Platforms:** Tableau, Power BI, Azure, BigQuery, Git/GitHub
- **Statistical Methods:** Statistical Forecasting, Time Series Analysis, Regression, ANOVA, GLMs, Hypothesis Testing

WORK EXPERIENCE

Navy Research Laboratory | Monterey, CA

Jun 2024 - Aug 2024

Research Intern - Data Analysis

- Executed comparative analysis on 100 km² of observational climate data, utilizing Xarray databases and operations to assess differences between WRF and NAAPS model outputs, directly informing regional climate model calibration initiatives.
- Quantified a 31% improvement in the accuracy of meteorological and aerosol forecasts in NAAPS-RA v2 vs. NAAPS-RA v1 climate models by applying RMSE metrics and time-series analysis, providing significant insights into climate models.
- Managed the full data lifecycle for 30+ GB of raw, gridded data with ETL methods using Python and Excel, and visualized critical findings derived from the cleaned dataset to highlight climate model performance improvements.
- Communicated strategic findings and climate-centered data insights to stakeholders via concise Quad Chart Reports.

Lawrence Berkeley National Lab | Berkeley, CA

Jun 2023 - Aug 2023

Research Intern - Data Analysis

- Engineered a structured dataset for geospatial and climate modelling by collecting and quality-checking over 300,000 environmental sensor readings (temperature, snow depth) from the East River Valley, Colorado, using Excel.
- Developed predictive models to forecast snowpack variability, achieving an 87% classification accuracy by correlating solar radiation and topographic orientation using Python to predict snowpack variability across high-altitude terrain.
- Co-authored a research manuscript and disseminated research findings in an LBNL-hosted webinar, using data visualizations to highlight important drivers of snowpack variability to 10+ researchers and facilitating future research collaboration.

ACADEMIC/PERSONAL PROJECTS

Machine Learning Final Project - An ML Approach to Rainfall Prediction | Davis, CA

Nov 2024 - Dec 2024

- Built machine learning models (Random Forest, XGBoost, Time Series) to forecast rainfall patterns across 100 monitoring sites in Northern California using 5 years of historical weather data.
- Cleaned and organized a multi-year time series dataset, engineered features (lag variables, precipitation trends), and trained models with Pytorch, TensorFlow, and scikit-learn.
- Assessed ML model performance using MAE and R² scores for clusters of sensors using K-Means Clustering, and found that the Random Forest model outperformed time series and XGBoost, improving prediction accuracy by 47%.
- Documented project results and findings in an organized GitHub repository that includes visualizations, model evaluations, and written analyses of each model's advantages and disadvantages.

Business Intelligence Dashboard - Scents and Shines Jewelry Store | Davis, CA

Jun 2025 - Sep 2025

- Designed and deployed a Power BI dashboard integrating sales, customer engagement, and inventory data.
- Identified product description optimizations that led to a 10% increase in store traffic within two months.
- Automated reporting pipelines, enabling the store owner to track real-time sales performance, coupon use, and product mix.
- Applied Power Query for data cleaning and preparation, and used DAX to build dynamic measures for store KPIs.